INCH-POUND

MIL-DTL-17/94G
w/AMENDMENT 1
06 February 2026
SUPERSEDING
MIL-DTL-17/94G
19 June 2015

MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, FLEXIBLE, COAXIAL, 75 OHMS, M17/94-RG179

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.

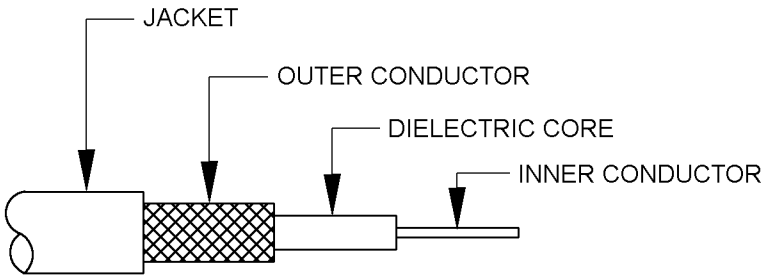


FIGURE 1. Configuration.

TABLE I. Description.

Components	Construction details
Inner conductor	Seven strands of silver-coated, annealed-copper-covered, steel wire, each strand, .004 inch diameter. Overall diameter: 0.012 inch ± 0.001
Dielectric core	Type F-1: Solid, extruded PTFE. Diameter: 0.063 inch ± 0.003.
Outer conductor	Single braid of AWG No. 38 silver-coated copper wire. Diameter: 0.084 inch, maximum. Coverage: 92.3%, nominal Carriers: 16 Ends: 5 Picks/inch: 12.0 ± 10%
Jacket	Type IX: FEP. Diameter: 0.100 inch ± 0.005.

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ENGINEERING INFORMATION:

Continuous working voltage: 900 V rms, maximum.
Operating frequency: 3 GHz, maximum.
Velocity of propagation: 69.5 percent, nominal.
Power rating: See figure 2.
Operating temperature range: -55° to +200°C.
Inner conductor properties:
DC resistance (maximum at 20°C): 24.45 ohms per 100 feet.
Elongation: 8 percent, minimum.
Tensile strength: 50 $klb_f/inch^2$, minimum.
Engineering note: This cable useful in general purpose, high temperature applications (see connector series “SMB” and “SMC” per MIL-PRF-39012. NATO preferred type MWR-33).

REQUIREMENTS:

Dimensions, configuration, and descriptions: See figure 1 and table I.

Environmental and mechanical:
Visual and mechanical examination:
Out-of-roundness: Not applicable.
Eccentricity: 10 percent, maximum.
Adhesion of conductors:
Inner conductor to core: 1.5 pounds, minimum; 4 pounds, maximum.
Aging stability: Not applicable.

Stress-crack resistance: +230°C ± 5°C.
Outer conductor integrity: Not applicable.
Cold bend: -55°C ± 2°C.
Dimensional stability: +200°C ± 5°C.
Inner conductor from core: .187 inch, maximum.
Inner conductor from jacket: .250 inch, maximum.
Contamination: Not applicable.
Bendability: Not applicable.
Flammability: Applicable.
Weight: 0.0108 pound per foot, maximum.

Electrical:

Continuity: Applicable.
Spark test: 2,000 V rms, + 10%, -0%.
Voltage withstanding: 2,000 V rms, +10%, -0%.
Insulation resistance: Not applicable.
Corona extinction voltage: 1,200 V rms, minimum.
Characteristic impedance: 75 ohms ± 3.
Attenuation: 21.0 dB per 100 feet, maximum, at 0.4 GHz.
Structural return loss: Not applicable.
Capacitance: 23.0 pF per foot, maximum.
Capacitance stability: Not applicable.
Capacitance unbalance: Not applicable.
Transmission unbalance: Not applicable.
Mechanically induced noise voltage: Not applicable.
Time delay: Not applicable.

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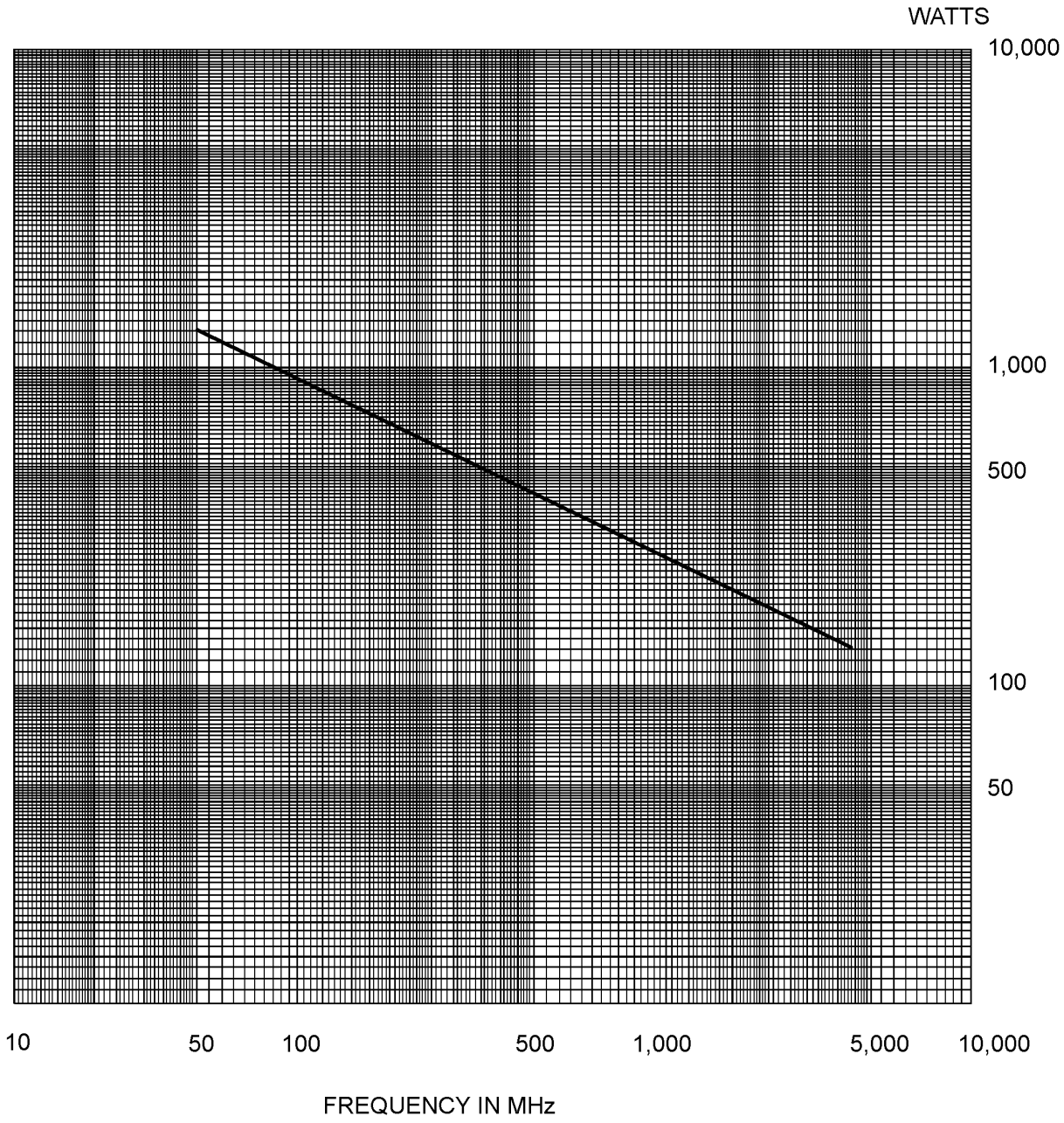


FIGURE 2. Power rating.

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Part or Identifying Number (PIN): M17/94-RG179.
Marking: Applicable
Marking Durability: 125 cycles (250 strokes), 1 pound
Workmanship: Applicable
Qualification: Shall consist of the inspections specified in table III.

Supersession data: See table II.

TABLE II. Cross reference of part number.

PIN	Superseded PIN or type designation
M17/94-RG179	RG-1798/U; RG-187A/U

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TABLE III. Qualification and Group A, B, and C Table.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
I	In-process inspection:		4.5	Entire sample		
	Continuity	3.7.1	4.8.2	Entire sample		
	Spark test	3.7.2	4.8.3	Entire sample		
	Voltage withstanding	3.7.3	4.8.4	Entire sample		
	Insulation resistance	N/A				
	Visual and mechanical inspection	3.6	4.8.1	Entire sample	Per Table VIII	A
	Diameter measurements.	3.6.1	4.8.1.1	Entire sample	Per Table VIII	A
	Out-of-roundness	N/A				
	Eccentricity of inner conductor	3.6.3	4.8.1.3	2	Per Table VIII	A
	Adhesion of conductors	3.6.4	4.8.1.4	4	Per Table VIII	A
	Marking	3.8	4.8.1	Entire sample	Per Table VIII	A
	Marking Durability	3.8.2	4.8.39	3	Per Table VIII	A
	Workmanship	3.10	4.8.1	Entire sample	Per Table VIII	A
II	Corona extinction voltage	3.7.5	4.8.6	1	Per Table VIII	B
	Characteristic impedance	3.7.6	4.8.7	1	Per Table VIII	A
	RF transmission loss (attenuation)	3.7.7	4.8.8	2	Per Table VIII	A
	Standing wave ratio (return loss)	N/A				
	Capacitance	3.7.9	4.8.10	1	Per Table VIII	B
	Capacitance stability	N/A	4.8.11	1	Per Table VIII	C
	Capacitance unbalance	N/A				
	Transmission unbalance	N/A				
	Inductance	N/A				
	Mechanically induced noise voltage	N/A				
	Time delay	N/A				
	Aging stability	N/A				

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TABLE III. Qualification and Group A, B, and C Table III – Continued.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
II	Stress crack resistance	3.7.16	4.8.17	4	Per Table VIII	C
	Outer conductor integrity	N/A				
	Cold bend	3.7.18	4.8.19	3	Per Table VIII	B
	Dimensional stability	3.7.19	4.8.20	1	Per Table VIII	C
	Contamination	N/A				
	Bendability	N/A				
	Flammability	3.7.22	4.8.23	1	Per Table VIII	C
	Flame propagation	N/A				
	Acid gas generation	N/A				
	Halogen content	N/A				
	Immersion	N/A				
	Smoke index	N/A				
	Toxicity index	N/A				
	Durometer hardness	N/A				
	Weathering	N/A				
	Abrasion resistance	N/A				
	Tear strength	N/A				
	Heat distortion	N/A				
	Physicals (aged)	N/A				
	Tensile strength and elongation	N/A				
	Weight	3.9	4.8.24	1	Per Table VIII	B

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Referenced documents. In addition to MIL-DTL-17, this document references the following:

MIL-PRF-39012

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 6145-2026-005)

Review activities:
Army – AT, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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